

Microsoft sues US government

Microsoft has sued US government over the order which allows cops to see customer email and records.
<http://dgit.in/microsueus>

**GMO Food**

The latest Genetically Modified Food White button mushroom is edited using CRISPR.
<http://dgit.in/CRISPRX>



Lifeblood for your system

Taking the human body inspired designs to a completely another level, Dr. Bruno Michel, IBM Research, Zurich tells us about his research and the development of a blood like technology for computers.



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How long has electronic blood been under development?

Bruno: The actual project name is called REPCOOL, which is short for “Redox flow electrochemistry for power delivery and cooling”. The project was awarded a grant from the Swiss National Science Foundation in September 2013 and in addition to IBM Research, other partners on the project include ETH Zurich, the Paul Scherrer Institute and the University of Lugano.

The general concept has been something I first began focusing on in 2006 when the Stern Reporter was released. This triggered me to pivot my research towards inventing incredibly powerful computers, which are densely packed and therefore very energy efficient using 3D chip stacks.

You can read about his vision in this paper:
<http://dgit.in/FDSscaling>

How does it do both energy delivery and cooling at the same time?

Bruno: This groundbreaking approach is modelled after the structure and power supply of the brain, where blood capillaries serve both to cool it and to supply it with energy. Compared to today's top computers, however, the human brain is roughly 10,000 times denser and 10,000

times more energy-efficient. The research team believes that their approach could reduce the size of a computer with a performance of 1 petaflop/s from the dimensions of a school classroom to that of an average PC, or in other words to a volume of about 10 liters.

To cool 3D chips efficiently, IBM scientists have already developed fluid-cooled test systems with capillary cooling structures that allow the fluid to flow between the individual chip layers, a process called “interlayer cooling” that achieves a cooling capacity of 3 kW/cm³.

The research team now takes this approach one step further by developing an integrated cooling and energy distribution system based on an electrochemical flow battery to address the power supply challenge in a sustainable way. Their approach is to distribute the energy electrochemically via what are known as redox chemicals in the capillary fluid instead of through electrical wires. Unlike conventional batteries, which store energy in solid electrode materials, a flow battery transports energy simply by circulating the redox chemicals. Thus the electrolytes are charged by an electrochemical cell called an electrolyzer and discharged via numerous distributed receiver electrodes on a single layer of a 3D chip stack. Large electrochemical flow batteries of this kind already exist for such purposes as grid-

scale electricity storage from renewable sources, but this technology must still be significantly improved and miniaturized for use in future chip stacks.

What is electronic blood made of? What would we call it, if we were to give it a technical name?

Bruno: The technology is called a Redox Flow Battery, which has been around for several decades and the liquid we are using is primarily vanadium.

While the primary suggested use for this is in data centers and supercomputers, do you think such a system can be used in consumer devices, like smartphones and tablets, in future?

Bruno: The research the team develops within the REPCOOL project will scale from supercomputers to cloud computers and wearable devices, but it's too early to make any predictions at this point.

What are the major challenges in the application of such a technology?

Bruno: To answer this question I would like to cite the following paper I co-authored titled - Integrated Microfluidic Power Generation and Cooling for Bright Silicon MPSoCs <http://dgit.in/IMPGCBMSM>

To allow a full electrochemical power supply of chip stacks from the same footprint a two pronged approach is